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Accurate Jitter Decomposition in High-Speed Links

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Abstract—**Jitter performance plays a crucial role in bit-error -rate of a high-speed digital communication system. Jitter decomposition is a key tool to accurately derive each type of jitter as well as total jitter in a system and identify the root causes of jitter. In this paper, we propose a jitter decomposition algorithm using least squares (LS) which simultaneously separates inter-symbol interference (ISI), random Jitter (RJ) and periodic Jitter (PJ). The algorithm includes a new time domain ISI model based on channel pulse response which is more effective than a conventional cursor convolution technique. The proposed jitter decomposition method is able to obtain the estimated individual jitter component value with great accuracy by using fewer samples of total jitter data compared with conventional methods. The simulation and hardware experiment demonstrate the efficiency and accuracy of the proposed method.**

Keywords—jitter decomposition; least squares; lossy channel; high-speed communication system; bit-error-rate

I. Introduction

With the ever-increasing demand of high speed data rate in serial communication systems, jitter becomes a dominant factor affecting system performance and the bit-error-rate (BER). As it also limits timing margin for the system today, accurate jitter analysis is crucial for next-generation I/O design in order to obtain an acceptable BER.

Jitter is defined as the deviation of transition edges from their ideal location in time and contains multiple components each with different characteristics. The total jitter (TJ) in data signal consists of deterministic jitter (DJ) and random Jitter (RJ). RJ follows unbounded Gaussian distribution due to noise sources (such as thermal noise, flicker noise, shot noise etc). DJ obeys bounded distribution and can be decomposed into periodic jitter (PJ) and data dependent jitter (DDJ), bounded uncorrelated Jitter (BUJ). PJ comes from external deterministic noise sources coupling into a system, such as switching power supply noises. DDJ is further divided into duty cycle distortion (DCD) and inter-symbol Interference (ISI). Non-idealities, such as asymmetric rising and falling edges of the clock path generate DCD which the duration of logical '1' is different from the duration of a logical '0'. ISI is caused by the bandwidth limitation, loss and reflection of the channel. DDJ is related to the bit sequence.

Understanding the amount of jitter introduced by each jitter source is imperative for predicting overall system performance [1]. Jitter decomposition is a key tool used in such scenarios to identify the root causes of jitter. Jitter can be measured using different methods either using external instrument (oscilloscope, spectrum analyzer and time interval analyzer (TIA)) or on-chip circuit jitter measurement. However, these on-chip jitter measurement circuits [2-6] require a large amount of die area if the jitter histograms have to be collected in real-time. Different off-chip decomposition approaches have been developed in the past. Generally, there are three popular category approaches to decompose jitter: 1) the ones based on histogram or statistical methods; 2) the ones with frequency-domain based analysis; and 3) the ones with time-domain based analysis relying on jitter measurements carried out in real-time.

The histogram methods are based on the popular Gaussian tail model [1] by using the probability distribution of collected jitter values. For example, RJ could be observed at the outer tails of a TJ distribution. Thus, this type of method is also referred to as a tail fitting algorithm. Researchers reported various methods to separate RJ and DJ components with different tail fitting algorithms [7-9]. However, a large amount of jitter samples is required to correctly identify the fit tail part of distribution. Deconvolution methods [10] rely on the idea that in histogram based analysis, a TJ probability density function (PDF) is given as convolution result of the RJ and DJ components. If one of these two components is approximately estimated, one can use a deconvolution algorithm to determine other components, and thus to retrieve the Gaussian model parameters. However, a major drawback of these methods is that they suffer from lack of accuracy because either the DJ or RJ component has to be estimated prior to the deconvolution.

For the frequency-based method, the time-domain series of jitter can also be represented and analyzed in the frequency domain using the Fourier transform (FT) [11-14]. Then, researchers can use the power spectral density (PSD) to represent the jitter spectrum by applying averaging techniques. Correspondingly, peaks in the spectrum can be interpreted as PJ or DDJ, while the average noise floor denotes the power of RJ. However, they used a clock pattern to estimate the RJ and PJ in the system and the ISI cannot be derived from long run-length patterns.

Jitter analysis techniques based on time-domain [15-17] rely on jitter measurements carried out in a real-time mode. This is only possible to those dedicated real-time measurement systems, such as high-speed sampling scopes or TIA. Those instruments also include the histogram or spectral test methods. Unfortunately, histogram, spectral and time-domain methods need sufficient memory depth to acquire enough data so that the accuracy can be assured through these digital signal processing (DSP) techniques [18]. In [19], a low-cost jitter separation method based on ADC testing was developed. However, this method is limited to separate jitter in only clock channel with low speeds.

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One important jitter component needed to be addressed is ISI jitter. The ISI jitter plays an important part in the TJ. The ISI modelling methods have been introduced in the past. In [12, 20], they modelled the channel as a first-order and a second-order low-pass filter. However, in presence of discontinuities, such a model is too simple to represent the real channel and becomes invalid. In [21], the ISI modelling is commonly based on the convolution technique. However, it is very time consuming since the cursor is usually 100-bit long.

In this paper, we present a jitter decomposition algorithm based on LS which has advantages of 1) accurate estimation for PJ, ISI and RJ; 2) fewer data samples than the instrument test which saves the memory requirement; and 3) the new ISI modelling is accurate and efficient to ISI jitter estimation. In this decomposition algorithm, the ISI jitter model based on channel pulse response in time domain is simpler than the conventional ISI cursors convolution technique. Another advantage of the proposed ISI modelling is its being more accurate and realistic than the low pass filter model. The PJ and RJ were also modelled by a traditional method [1]. The LS was used to obtain parameter values in the above-mentioned jitter model [22-23]. Compared with the simulation and conventional instrument test methods, the proposed method shows great accuracy in the jitter decomposition.

The rest of the paper is organized as follows: In Section II, the conventional PJ, RJ models are reviewed and the new ISI jitter model is presented. In Section III, the TJ model is introduced. A LS method is applied to the TJ model and the estimation of jitter component is derived. In Section IV, the validation for the ISI model and jitter decomposition simulation results are presented. In Section V, the hardware experiment is presented. Section VI concludes the paper.

II. Jitter Modelling

This section presents PJ, RJ and ISI jitter models considered in the algorithm. PJ and RJ models are widely used in popular jitter analysis. Specifically, we developed a new time domain ISI jitter model for a lossy channel which is efficient at ISI estimation.

A. Periodic jitter(PJ)

PJ is a repeating jitter signal at certain frequencies. It is typically derived from the noise in a switching power supply or caused by PLL reference clock feedthrough. The mathematical model of PJ [1] is

$$\begin{aligned}\Delta t_{PJ}[n] &= A\sin(2\pi f_0\,(t-nT)+\emptyset)\\ &= a\sin\left(\frac{2\pi f_0 n}{f_s}\right)+b\cos\left(\frac{2\pi f_0 n}{f_s}\right)\end{aligned} \quad (1)$$

Where $\Delta t_{PJ}[n]$ is a jitter amount at sampling time nT; A is the amplitude of PJ; f_0 is the frequency of PJ. In a real system, it can be input reference clock of PLL or power supply noise and extracted from the data through spectral analysis. In this paper, we consider the PJ from reference clock as an example; f_s is frequency of data stream, and $\emptyset$ is the phase.

a and b are the estimation parameters for PJ in the algorithm.

B. Random jitter (RJ) modelling

RJ is caused by unbounded jitter sources, such as thermal noise, flicker noise, and shot noise which can be modeled as Gaussian white noise. Gaussian jitter PDF is defined as

$$f_{RJ}(\Delta t)=\frac{1}{\sqrt{2\pi\sigma}}e^{-\frac{(\Delta t-\mu)^2}{2\sigma^2}} \quad (2)$$

μ is the mean and σ is the standard deviation of Gaussian distribution. In this model, the μ is zero and σ is the estimation parameter for RJ in the algorithm.

C. Inter-symbol interference jitter (ISIJ) modelling

ISI is caused by reflections and loss in a channel. Fig. 1 shows a single-bit response after a lossy channel. The pulse becomes widened and attenuated, and it occupies the pre-cursor and post cursor samples. Traditionally, ISI cursors convolution technique is used in ISI modelling. However, this process is time-consuming when the number of cursor is large. For instance, typical single-bit responses are often more than 100bit long [21]. This convolution approach fails when the non-linearity of the system is severe, or when the data pattern is non-white.

Since sophisticated methods, like transmitter finite impulse response (TX-FIR) equalizer, are needed to properly equalize to the precursor, we consider the k-bit post cursor has a dominant effect on ISI. We use an example to illustrate the time domain ISI model. Considering the data sequence b1-b6 as shown in Fig.2, the black curve is the ideal data sequence and blue curve is the actual data sequence due to ISI. b6 is the current bit, b1-b5 are preceding five bits (5 bits post cursor). b1-b5 has 32 binary combinations. If b1-b5 is 01011 as shown in Fig.2 a), the time deviation of actual b6 edge and the ideal b6 edge is defined as ISI induced jitter J_{11}. The index 11 is the decimal representation of 01011. If b1-b5 is 01101 shown in Fig.2 b), the time deviation of actual b6 edge and the ideal b6 edge is thus noted as jitter J_{13}. The index 13 is the decimal representation of 01101. Different b1-b5 binary combinations generate different ISI jitter amount to current bit b6. Table I describes the ISI model parameters which include binary combinations, a corresponding jitter J and a sign C. In this example, the 5-bit post cursor has 32 binary combinations and each binary combination generates a corresponding jitter value J. The corresponding sign C is used to represent which kind of binary combination of 5-bit post cursor is selected. In Fig.2 a), the 5-bit binary combination is 01011 and the corresponding jitter value is J_{11}, the corresponding sign C_{11} is equal to 1 while other corresponding signs are all zeros.

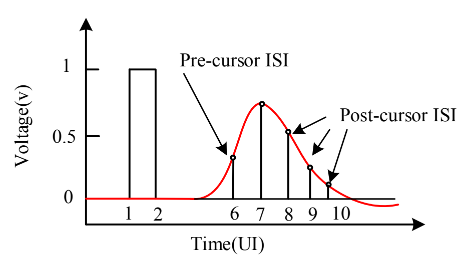

Fig. 1. The pulse response of the channel

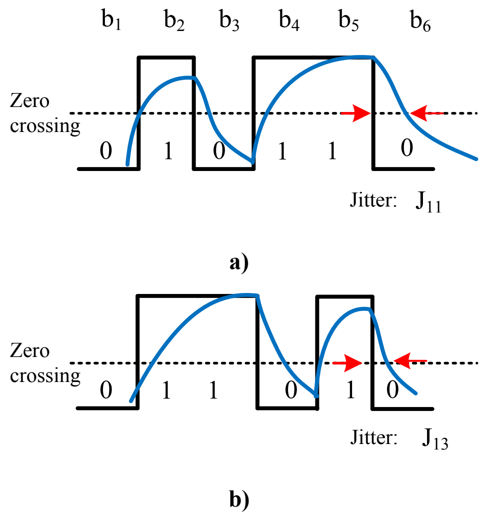

Fig. 2. ISI jitter modelling (a) and (b)

Based on the observation that different post cursor binary combinations generate different ISI jitter amounts on the current bit, we can use a formula to describe the ISI jitter model. The equation is given by

$$\Delta t_{ISI}[n] = f(\, b_{n-k} b_{n-k+1} \dots b_{n-1}) = \sum_{l=0}^{2^k-1} (J_l \times C_l[n]) \quad (3)$$

$$C_l = \begin{cases} 1, \text{ if } \mathit{Binary\ to\ decimal}\ (\, b_{n-\mathrm{k}} b_{n-\mathrm{k}+1} \dots b_{n-1}) == l \\ 0, \text{if } \mathit{Binary\ to\ decimal}\ (\, b_{n-\mathrm{k}} b_{n-\mathrm{k}+1} \dots b_{n-1})\ != l \end{cases}$$

where $\Delta t_{ISI}[n]$ represents ISI jitter of bit n at sample time nT. It is a function of k bits post cursor from b_{n-k} to b_{n-1} which has 2^k binary combinations. l is the decimal representation of binary combination $b_{n-\mathrm{k}} b_{n-\mathrm{k}+1} \dots b_{n-1}$. J_l is the jitter amount of the lth binary combination to the current bit b_n. C_l is a corresponding sign to represent the binary combination of the k-bit post cursor. The equation means ISI jitter of the current b_n is the jitter amount of the k bit post cursor. J_l is the estimation parameter in the algorithm. The computation is simpler than cursors convolution.

The post cursor number k can be obtained from the channel pulse response given the threshold voltage as shown in Fig.1.

III. Jitter Decomposition Algorithm

In this section, we present the TJ modelling used in this study, and then explain the details of the decomposition algorithm.

TABLE I. THE PARAMETER OF ISI JITTER MODEL

b1-b5 binary combinations	Corresponding Jitter	Corresponding sign
00000	J_0	C_0
…	…	…
01011	J_{11}	C_{11}
…	…	…
11111	J_{31}	C_{31}

A. Total jitter (TJ)

The TJ in time domain is considered as the linear sum of DJ components and the square root of the RJ components [21].

With the proposed ISI jitter modelling, the TJ in bit n with PJ, RJ and ISI can be simulated as:

$$\begin{aligned} x[n] &= d_n \times \left[\Delta t_{PJ}[n] + \Delta t_{ISI}[n] + \Delta t_{RJ}[n]\right] \\ &= d_n \times [a \sin\left(\tfrac{2\pi f_0 n}{f_s}\right) + b \cos \tfrac{2\pi f_0 n}{f_s} \\ &\quad + \textstyle\sum_{l=0}^{2^k-1} (J_l \times C_{ln}) \;+ \delta[n]] \end{aligned} \quad (4)$$

where $x[n]$ represents the TJ amount at sampling time nT. Parameters a, b, f_s, f_0, C_l, J_l have been described in previous section, δ[n] is pure RJ in sampling time instance nT. d_n is a binary value indicating the existence of a data transition from bit $n-1$ to bit. $d_n = 1$ only when a falling or rising edge presents from bit $n-1$ to bit n. $d_n = 0$ when there is no data transition from bit $n-1$ to bit n. For example shown in Fig.3, $d_n = 1$ when a rising edge presents in the data stream from b_{n-1} to b_n, while $d_n = 0$ when there is no data transition from b_{n-1} to b_n. That means no jitter exists in this bit.

B. Jitter decompostion by Least Squares (LS)

In order to estimate the parameters in the proposed model, a LS based decomposition method is proposed. Equation (4) shows that TJ is a linear equation. For a linear time invariant system, LS estimation overcomes the convergence problem [22-23] and does not require any special distribution properties for the input. Based on this, we applied the LS to estimate the PJ, RJ, and ISI parameters $[a \;\; b \;\; J_0 \;\; J_1 \cdots J_{2^k-1}]$.

In order to obtain the estimation of PJ, ISI and RJ, we define that the length of a TJ sequence is i. The TJ in each bit is $x[1], x[2], \dots, x[i]$ taken at sampling time $1T, 2T, \dots iT$, respectively. Then the total jitter sequence Z_i can be expressed by the following equation [22]

$$Z_i = \begin{bmatrix} x[1] \\ x[2] \\ \vdots \\ x[i] \end{bmatrix} = H_i \theta + V_i \quad (5)$$

where V_i is referred to the RJ vector (denoted as $[\delta[1]\, \delta[2] \dots \delta[i]\,]^T$) for the jitter sequence, θ = $[a \;\; b \;\; J_0 \;\; J_1 \cdots J_{2^k-1}]^T$ which is the estimation parameters for

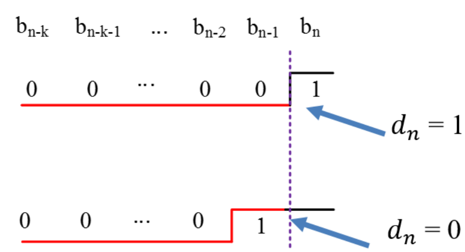

Fig. 3. The data transition sign d_n

PJ and ISI. H_i is the coefficient matrix for the whole jitter sequence. In (6), the first and second columns are PJ coefficients matrix, the third and the remainder columns are ISI coefficients matrixes

$$H_i = \begin{bmatrix} d_1 \times \sin\left(\frac{2\pi f_0}{f_s} \times 1\right) & d_1 \times \cos(\frac{2\pi f_0}{f_s} \times 1) & d_1 \times C_{01} & \cdots & d_1 \times C_{(2^k-1)1} \\ d_2 \times \sin\left(\frac{2\pi f_0}{f_s} \times 2\right) & d_2 \times \cos\left(\frac{2\pi f_0}{f_s} \times 2\right) & d_2 \times C_{02} & \cdots & d_2 \times C_{(2^k-1)2} \\ \vdots & \vdots & \vdots & \vdots & \vdots \\ d_i \times \sin\left(\frac{2\pi f_0}{f_s} \times i\right) & d_i \times \cos\left(\frac{2\pi f_0}{f_s} \times i\right) & d_i \times C_{0i} & \cdots & d_i \times C_{(2^k-1)i} \end{bmatrix} \quad (6)$$

In (6), C_{li} can be extracted from the data stream. For example, if a binary combination of post cursor of the ith bit data stream is 01101, then C_{12i} is 1 and the other C_{xi}s are 0. f_s, and f_0 are known parameters from the previously-mentioned section.

The solution $\hat{\theta}_i$ for length of jitter sequence i is

$$\hat{\theta} = [H_i^T H_i]^{-1} H_i^T Z_i \quad (7)$$

The estimation parameter $\hat{\theta}$ is

$$\hat{\theta} = [\hat{a} \;\; \hat{b} \;\; \hat{J}_0 \;\; \hat{J}_1 \; \cdots \; \hat{J}_{2^k-1} \;]^T \quad (8)$$

With the estimation results, the amplitude of the PJ is

$$\hat{A} = \sqrt{\hat{a}^2 + \hat{b}^2} \quad (9)$$

To calculate peak-to-peak ISI, we can find the rising edge and falling edge of ISI from $\{ \hat{J}_0 \;\; \hat{J}_1 \; \cdots \; \hat{J}_{2^k-1}\}$. Namely,

$$\widehat{ISI}_{rising} = max_{rising} \;(\hat{J}_0\, \hat{J}_1 \cdots \hat{J}_{2^k-1}) - min_{rising} \;(\hat{J}_0\, \hat{J}_1 \cdots \hat{J}_{2^k-1})$$

$$\widehat{ISI}_{falling} = max_{falling} \;(\hat{J}_0\, \hat{J}_1 \cdots \hat{J}_{2^k-1}) - min_{falling} \;(\hat{J}_0\, \hat{J}_1 \cdots \hat{J}_{2^k-1})$$

$$\widehat{ISI}_{pk-pk} = 0.5 \times \left(\widehat{ISI}_{rising} + \widehat{ISI}_{falling}\right) \quad (10)$$

The variance of RJ is

$$\sigma^2 = var\left(Z_i - H_i \hat{\theta}\right) \quad (11)$$

The flow chart of the proposed algorithm is given in Fig. 4.

The jitter sequence with PJ, ISI, and RJ is the input of the algorithm. The ISI matrix is formed based on data stream information and the PJ matrix is formed based on the data rate and PJ frequency (from the spectral analysis). According to (7), the initial estimation jitter component value $\hat{\theta}$ could be obtained by LS. The final estimation of PJ, ISI, and RJ values is based on (9), (10), (11), respectively.

IV. SIMULATION RESULT

In this section, the proposed new ISI modelling and decomposition methods are validated by our Matlab simulation. The pseudo-random binary sequence (PRBS7) data length is 1.27k bits and the data rate is 10Gb/s in the simulation.

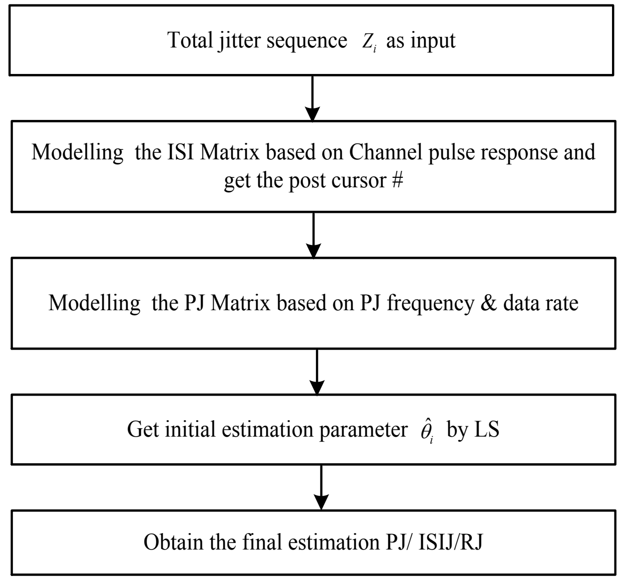

Fig. 4 The flow chart of proposed algorithm

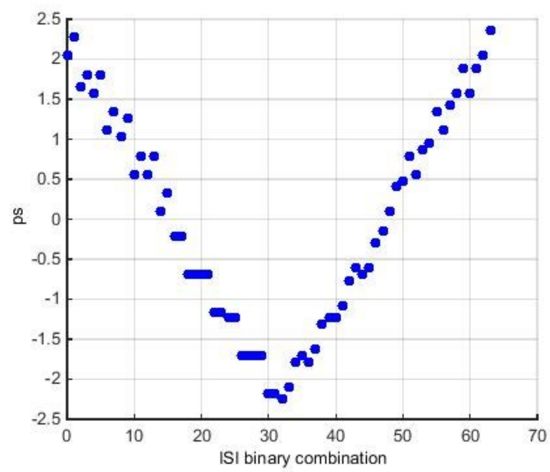

Fig.5 The classification of ISI jitter

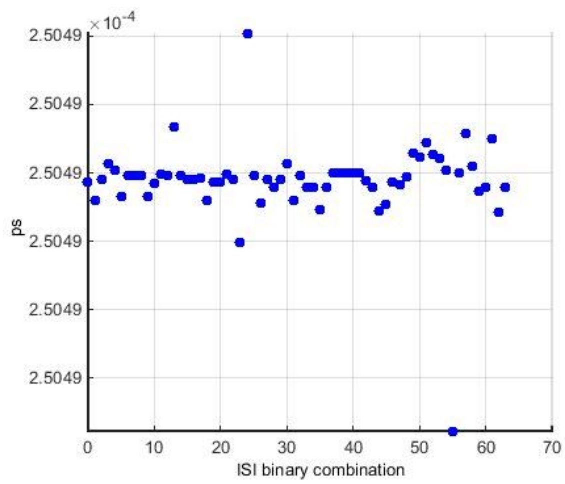

Fig.6 The error of estimated jitter and the added jitter

A. Validation of the ISI jitter modelling

The extraction of S-parameter of a PCB transmission line with insertion loss 3dB at 5Ghz was used to generate the ISI jitter sequence. In order to verify that the ISI jitter is k-bit post cursor dependent, we classified the ISI jitter sequence to 2^k binary combinations. The post cursor number k of the transmission line is 6, which was obtained from the channel pulse response. These 6-bit post cursors have 64 binary combinations from 000000 to 111111. The corresponding jitter amount are from J_0 to J_{63}. The classification result in Fig.5 shows different ISI jitter amounts of corresponding binary combinations. The peak-peak amount of ISI is about 4.6 ps according to (9). We consider that the added ISI jitter is 4.6 ps as the reference ISI jitter in Part A and B of this section.

In order to verify the decomposition algorithm, the ISI jitter sequence is sent to the algorithm. The comparison of the estimated ISI jitter and added jitter is plotted in Fig.6. It shows that the difference of the estimated jitter and added jitter is close to 0 ps.

B. Validation of PJ, RJ and ISI jitter

In the simulation, different amounts of TJ sequences were generated by Matlab Simulink toolbox. The TJ sequence with different PJ (0ps, 20ps, peak-peak value), RJ (0ps, 2.13ps, rms value), and ISI jitter (4.6ps, peak-peak value) were sent to the algorithm. The PJ was a sine wave with 100MHz frequency. The simulation results are summarized in Table II. The estimation error of ISI is less than 0.6 ps. The estimation error of PJ and RJ is close to 0 ps.

V. MEASURMENT

To verify the ISI modelling and the accuracy and efficiency of the algorithm, a hardware test bench was used (shown in Fig.7) to measure the jitter components. A Tektronix BSA286C BERTscope was used to generate data stream with PJ and RJ. An Agilent Infiniium Wide-Bandwidth Oscilloscope was used to measure the jitter with internal software. The same PCB transmission line in simulation part A was used to generate the ISI jitter. All experiments were done at a data rate of 10Gb/s.

A. Validation of the ISI jitter modelling

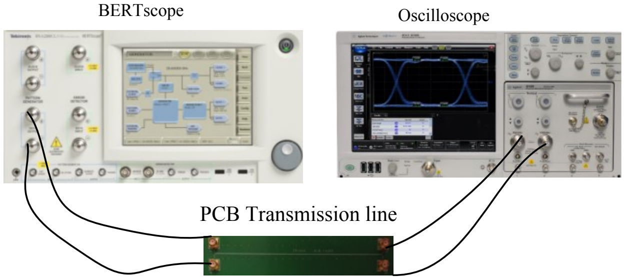

Fig.7 The experimental test bench

Since the ISI jitter (peak-peak) generated by the PCB transmission line is unknown, we used the oscilloscope to measure the ISI jitter as a reference. In the experiment, PRBS7 without jitter generated by BERTscope was sent to the PCB transmission line and the output of transmission line was connected to oscilloscope to obtain the ISI jitter value.

In the simulation, the TJ sequences only with ISI jitter were sent to the algorithm and the estimation ISI value was obtained. The comparison is listed in Table III. In the added jitter term, the unknown for ISI means that the ISI jitter amount is unknown when the transmission line is added. It shows that the ISI value in the measurement is 5.1 ps and the one in the simulation is 4.6 ps. Both values are very close. Therefore, the ISI modelling is validated. RJ (rms value) in the simulation is 0ps while the one in the measurement is 1.4ps due to the instrument noise. PJ in the proposed method is 0.07ps (peak-peak) while the oscilloscope result is about 1.4ps.

B. Comparison of the accuracy and the sample data

In the experiment, PRBS7 with different PJ (0ps, 20ps, peak-peak), RJ (0ps, 2.13ps, rms) generated by BERTscope was sent to the PCB transmission line. The same amount of PJ, RJ, and ISI were added to the TJ sequence in the simulation.

Table IV shows the estimation results and the sample data in the simulation. Table V shows the instrument test result and the sample data number. When the added ISI is 0, the oscilloscope result of ISI jitter is about 3.2 ps due to the cable while in the proposed algorithm the estimation is about 0 ps. The error of added and estimated PJ is less than 0.1 ps. The error of added and estimated RJ is less than 0.03 ps. While in the instrument test, the maximum error of added PJ and estimated PJ is about 2.4ps and the maximum error of added and estimated RJ is about 1.41ps. For all cases, the simulation results are more accurate than the ones from the instrument test. The data sample in each simulation is only 1.27k, but the instrument requires at least 200k data in the experiment with less accuracy. Therefore, the proposed algorithm has better accuracy using fewer data samples.

TABLE II. THE SIMULATIN RESULT OF PJ, RJ AND ISI

Added jitter(ps)			Simulation result (ps)		
RJ	***PJ***	***ISI(pk-pk)***	***RJ***	***PJ***	***ISI***
0	20	4.6	0.02	20.00	4.6
2.13	0	4.6	2.13	0.08	5.2
2.13	20	4.6	2.12	20.04	4.6

TABLE III. COMPARISION THE PROPOSED METHOD AND INSTRUMENT FOR ISI

Added jitter (ps)			The proposed method (ps)			Oscilloscope (ps)		
RJ	***PJ***	***ISI***	***RJ***	***PJ***	***ISI***	***RJ***	***PJ***	***ISI***
0	0	unknown	0	0.07	4.6	1.41	1.4	5.1

TABLE IV. THE PROPOSED METHOD RESULT OF PJ,RJ AND ISI

Added jitter(ps)			The proposed method result (ps)			
RJ	*PJ*	*ISI*	*Sample data(bit)*	*RJ*	*PJ*	*ISI*
0	20	0	1.27k	0.03	20.10	0
0	20	Unknown	1.27k	0.03	20.00	4.58
2.13	20	0	1.27k	2.12	20.04	0.83
2.13	20	Unknown	1.27k	2.12	20.04	4.59

TABLE V. INSTRUMENT RESULT OF PJ,RJ AND ISI

Added jitter(ps)			Oscilloscope result (ps)			
RJ	*PJ*	*ISI*	*Sample data(bit)*	*RJ*	*PJ*	*ISI*
0	20	0	600k	1.41	20.5	3.2
0	20	Unknown	200k	1.51	20.3	6.5
2.13	20	0	300k	2.9	17.6	3.2
2.13	20	Unknown	300k	2.49	19.8	6

VI. CONCLUSION

An efficient and accurate algorithm that simultaneously extracts the periodic jitter, random jitter and ISI jitter is presented. This method is based on time-domain ISI modelling which is simpler than the conventional cursor convolution technique. It utilizes fewer sample data while maintaining great estimation accuracy in both clock pattern and data pattern. The comparison between simulation results and the hardware test ones demonstrates the accuracy of the proposed jitter measurement method. However, there are still limitations in the proposed method. The proposed method cannot decompose the DCD and BUJ. It is only applied in low lossy channels, which means the eye diagram should be open before the receiver of system and the data logic value can be correctly determined. The algorithm needs the TJ sequences as an input which requires extra instruments such as TIA to store the jitter sequence.

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